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Antibiotics for Surgical Prophylaxis in Adult Patients

General Considerations

- Not all procedures require antibiotic prophylaxis. (For example, typically not required for clean procedures or most endoscopic procedures).
- Antibiotics **must be started** within 60 minutes before surgery (120 minutes for vancomycin or ciprofloxacin) and at least 50% should be infused prior to incision to be effective.
- Intraoperative redosing may be required for long cases (>2 h); see Table 3.
- Patients already receiving broad-spectrum beta-lactams (e.g. piperacillin-tazobactam, cefepime, ceftriaxone) for treatment of infection often do not require an additional pre-op prophylaxis antibiotic as long as the antibiotic has a similar spectrum of activity (e.g. patient on ceftriaxone MAY still need a pre-op dose of vancomycin if at risk for MRSA, but would NOT require a dose of cefazolin). Consider holding the standing dose until 1 hour prior to incision.
- Antibiotics during surgery can prevent infections. Post-procedure doses generally are NOT recommended. Extending systemic antibiotics beyond 24 hours after surgery is rarely of benefit. Patients with open wounds or drains in place do not benefit from more prolonged courses of antibiotic prophylaxis. Rare exceptions are approved in certain surgical site infection prophylaxis bundles, including solid organ transplant and ventricular assist device implantation.
- Alternate regimens are for patients allergic to or intolerant of the primary regimen. Patients with penicillin allergy are generally encouraged to use cephalosporins. Benefits of first-line therapy usually outweigh the very low risk of cross-reactivity. For more information, refer to the UR guideline on [Evaluation of Beta Lactam Allergy](#) and [Peri-Operative Antibiotic Prophylaxis in Patients with a History of Beta-lactam Reactions](#).
- These recommendations present reasonable, but not exclusive, choices for surgical prophylaxis. Other antibiotic choices may be appropriate based upon current or prior antibiotic therapies in a particular patient.
- These guidelines comply with all recommendations for surgical prophylaxis in the Surgical Care Improvement Project Quality Measures and are reviewed as these are updated.

Table 1. Recommended regimens by surgery type

Surgery	Recommended Regimen	Alternative Regimen if Allergy to First Line Regimen	Details
Cardiac Surgery – Refer to the Cardiac Surgery SSI Prevention Bundle - Adult			
Low risk MRSA ^a	One of the following: - Cefazolin + Gentamicin - Cefazolin + Ciprofloxacin	One of the following: - Vancomycin + Gentamicin - Vancomycin + Ciprofloxacin	Includes patients undergoing VAD placement Tobramycin in place of gentamicin if undergoing durable LVAD placement AND current femoral mechanical circulatory support (Impella, IABP, ECMO)
High risk MRSA ^a	One of the following: - Cefazolin + Vancomycin + Gentamicin - Cefazolin + Vancomycin + Ciprofloxacin	One of the following: - Vancomycin + Gentamicin - Vancomycin + Ciprofloxacin	
Orthotopic heart transplant	Vancomycin + Cefepime	Vancomycin + Ciprofloxacin	
Gastrointestinal Procedures			
Esophageal Gastroduodenal Biliary tract Other laparoscopic procedures)– only for high risk ^b	One of the following: - Ampicillin/ Sulbactam for cholangitis, cholecystitis, biliary obstruction - Cefazolin	One of the following: - Clindamycin + either: Gentamicin or Ciprofloxacin - Vancomycin + either: Gentamicin or Ciprofloxacin	Low risk patients and procedures do not require prophylaxis
Gastric bypass	One of the following: - Ampicillin/ Sulbactam - Cefazolin +/- Metronidazole	One of the following: - Clindamycin + either: Gentamicin or Ciprofloxacin - Vancomycin + Metronidazole + either: Gentamicin or Ciprofloxacin	
PEG tube placement	Cefazolin	Vancomycin or Clindamycin	
Colorectal	Optional Oral Gut Regimen ^c + one of the following:	Optional Oral Gut Regimen ^c + one of the following: Clindamycin +	Ertapenem, while effective, increases risk of antimicrobial resistance

	• Ampicillin/Sulbactam • Cefazolin + Metronidazole Alternative: • Ertapenem	either: Gentamicin or Ciprofloxacin, • Metronidazole + either: Gentamicin or Ciprofloxacin	
Appendectomy	Cefazolin + Metronidazole	Ertapenem	If ruptured, treat as infected
Cholecystectomy	Cefazolin	Ertapenem	Add vancomycin for high-risk patients ^d If incarcerated or strangulated bowel add metronidazole
Hernia repair	Cefazolin	Vancomycin	
Hepatectomy	Cefazolin If biliary involvement: Piperacillin/Tazobactam	Clindamycin + Gentamicin If biliary involvement: Vancomycin + Ciprofloxacin	
Distal Pancreatectomy	Cefazolin If biliary involvement: Piperacillin/Tazobactam	Clindamycin + Gentamicin If biliary involvement: Vancomycin + Ciprofloxacin	
Head & Neck Procedures			
Adenoidectomy, parotidectomy, thyroidectomy, tonsillectomy (clean without prosthesis)	Prophylaxis not recommended	Prophylaxis not recommended	
All others (clean with prosthesis, clean-contaminated, or contaminated)	One of the following: • Cefazolin + Metronidazole • Ampicillin/Sulbactam	One of the following: • Clindamycin + either: Gentamicin or Ciprofloxacin, • Metronidazole + either: Gentamicin or Ciprofloxacin	Cefazolin alone may be considered for certain clean procedures
Neurosurgery			
Craniotomy, laminectomy, spinal fusion, VP shunt	Cefazolin ^d	Vancomycin or Clindamycin	Add vancomycin for high-risk patients ^d
Obstetrics & Gynecology			
Cesarean section Sub-total hysterectomy	Cefazolin +/- Azithromycin	One of the following: • Clindamycin +	Azithromycin 500 mg IV once for rupture of membranes and/or

		Gentamicin +/- Azithromycin • Vancomycin + Gentamicin +/- Azithromycin	laboring prior to C-section Patients with intra-amniotic infection undergoing Cesarean delivery: Refer to Guidelines for Prophylaxis and Treatment of Peripartum Infections
Abdominal/vaginal hysterectomy OR Elective, high-risk laparoscopic procedures	One of the following: • Cefazolin + Metronidazole • Ampicillin/Sulbactam • Cefoxitin	One of the following: • Clindamycin + either: Gentamicin, Ciprofloxacin, or Aztreonam • Metronidazole + either: Gentamicin or Ciprofloxacin	
Unplanned D&C	Doxycycline 200 mg 1 hour prior to procedure		
Orthopedic			
Clean operations without implants involving hand, knee, or foot, arthroscopy	Prophylaxis not recommended	Prophylaxis not recommended	
Total joint replacement	Cefazolin ^d	Vancomycin or Clindamycin	Add vancomycin for high-risk patients ^d
Other orthopedic surgery	Cefazolin ^d	Vancomycin or Clindamycin	
Plastic Surgery and Breast Procedures			
Skin grafts, fat grafts, implants, complex reconstructions, mesh, mastectomy	Cefazolin ^d	Vancomycin or Clindamycin	Not required in most cases without implants or mesh; Add vancomycin for high-risk patients ^d
Thoracic Surgery			
All	Cefazolin	Vancomycin or Clindamycin	
Urological Procedures			
TURP, resection of bladder tumor	Cefazolin	Gentamicin	Regimen should be tailored to known microbiology data, if available
Other clean, clean-contaminated	Cefazolin	Vancomycin or Clindamycin	

procedures			
Penile or other prosthesis	Cefazolin + Gentamicin	Vancomycin or Clindamycin + Gentamicin	
Radical prostatectomy	Cefazolin	Vancomycin or Clindamycin	
Vascular Surgery			
Carotid and brachiocephalic procedures without prosthetic graft	Prophylaxis not recommended	Prophylaxis not recommended	
Dialysis access procedures	Cefazolin ^d	Vancomycin	Add vancomycin for high-risk patients ^d
Other vascular surgeries	Cefazolin ^d	Vancomycin	

^aCardiac surgical high risk for MRSA: history of MRSA infection, MRSA-screen positive, hemodialysis, patient resides in chronic care facility

^bHigh risk: Over 70 years old, acute cholecystitis, obstructive jaundice or common duct stones, emergency procedure, open cholecystectomy

^cOral gut regimen option: neomycin sulfate 1 g AND (erythromycin base 1 g **OR** metronidazole 1 g PO); give 3 doses over 10 hours after mechanical bowel prep on day before surgery or within 24hrs prior to surgery.

^dIn procedures involving placement of implants or foreign material, vancomycin is recommended in addition to cefazolin for patients known to be at high risk for MRSA (history of MRSA infection, MRSA-screen positive, hemodialysis, patient resides in chronic care facility).

Table 2. Recommended regimens by procedure type

Surgery	Recommended Regimen	Alternative Regimen if Allergy to First Line Regimen	Details
Gastrointestinal^e			
EGD/colonoscopy/sigmoidoscopy	Prophylaxis not recommended	Prophylaxis not recommended	
ERCP with bile duct obstruction and complete drainage in absence of cholangitis	Prophylaxis not recommended	Prophylaxis not recommended	With incomplete drainage, Cefazolin +/- Metronidazole
ERCP other than biliary obstruction	Prophylaxis not recommended	Prophylaxis not recommended	
EUS-FNA (mediastinal/pancreatic cysts)	Cefazolin	One of the following: • Vancomycin + Gentamicin	

		• Clindamycin + Gentamicin	
Variceal band ligation with active bleed	Cefazolin	Vancomycin or Clindamycin	
Cardiac			
Pacemaker or ICD insertion	Cefazolin ^f	Vancomycin or Clindamycin	Add vancomycin for high-risk patients ^f
Vascular			
Venous/arterial procedures (e.g. angiography, angioplasty, stent placement)	Prophylaxis not recommended	Prophylaxis not recommended	
Tunneled catheters	Prophylaxis not recommended	Prophylaxis not recommended	
Implantable access port	Cefazolin	Vancomycin or Clindamycin	
Neurosurgery			
Implantation of intrathecal pump	Cefazolin	Vancomycin or Clindamycin	
Urologic			
Nephrostomy tube placement, tube exchange, ureteral stent	Cefazolin	One of the following: • Vancomycin + Gentamicin • Clindamycin + Gentamicin	Regimen should be tailored to known microbiology data, if available
Interventional Radiology			
Biliary/GI embolization or chemoembolization, percutaneous liver ablation (recent surgery, instrumentation)	Cefazolin + Metronidazole	One of the following: • Vancomycin + Gentamicin • Clindamycin + Gentamicin	
PEG tube placement	Cefazolin	Vancomycin or Clindamycin	
Uterine artery embolization, liver/lung/renal ablation; vascular malformation embolization ^g	Prophylaxis not recommended	Prophylaxis not recommended	
TIPS procedures	One of the following:	One of the following:	

	• Cefazolin + Metronidazole • Cefoxitin • Ampicillin/Sulbactam	• Vancomycin + Gentamicin • Clindamycin + Gentamicin	
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^eMost endoscopic GI procedures do NOT require antibiotic prophylaxis for prevention of infective endocarditis.

^fCardiac surgical high risk for MRSA: history of MRSA infection, MRSA-screen positive, hemodialysis, patient resides in chronic care facility

^gPatients with necrotic tissue undergoing vascular graft should receive prophylaxis with cefazolin.

Table 3. Drug doses and redosing intervals

Drug	Dose	Redosing Interval (hours) ^h	Maximum Daily Dose ^h
Ampicillin/Sulbactam	3 g IV	2	18 g
Aztreonam	2 g IV	4	8 g
Cefazolin	2 g IV for patients < 120 kg 3 g IV for patients ≥ 120 kg	4	6 g (<120 kg) 9 g (≥120 kg)
Cefepime	2 g IV	4	6 g
Cefoxitin	2 g IV	2	8 g
Ciprofloxacin	400 mg IV	8	1200 mg
Clindamycin	900 mg IV	6	2700 mg
Ertapenem	1 g IV	24	1 g
Gentamicin ⁱ	5 mg/kg IV (see attached for dosing bands)	24	NA
Metronidazole	500 mg IV	8	1500 mg
Piperacillin/Tazobactam	3.375 g IV	2	13.5 g
Vancomycin	1 g IV over 1 h for patients < 80 kg 1.5 g IV over 1.5 h for patients ≥ 80 kg	12	2 g (< 80 kg) 3 g (≥ 80 kg)

^h Redosing recommendations are for patients with normal renal function. May consider additional redosing during procedures with excessive blood loss (1500 mL).

ⁱ Dosing should be limited to a single dose preoperatively and should be based on the patient's actual body weight unless patient is ≥20% over ideal body weight, then adjusted body weight should be utilized ($\text{AdjBW} = \text{IBW} + [0.4(\text{ABW} - \text{IBW})]$). If on dialysis or creatinine clearance < 10mL/min, use 2 mg/kg once.

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Key terms: Prophylactic, antibiotics, Surgery, prophylactic antibiotics for surgery

Attachments

 [Gentamicin dose bands.pdf](#)

Approval Signatures

Step Description

Approver

Date

Final Approval to Post	Elizabeth Conderman	2/11/2026
Clinical Council	Elizabeth Conderman	1/13/2026
Therapeutics Committee	Travis Dick	12/23/2025
Policy Owner	Stephanie Shulder	12/16/2025

Applicability

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